

Zheng Zhang

The Hong Kong Polytechnic University
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 Google Scholar

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RESEARCH INTERESTS

Medical image reconstruction, Brain-Computer Interfaces, Generative Models, Foundation Models for Healthcare.

EDUCATION

The Hong Kong Polytechnic University

2025 – present

Ph.D. Student

Advisor: Prof. Jing Qin

Imperial College London

2021 – 2022

M.Sc. in Biomedical Engineering

Advisor: Prof. Mengxing Tang

University of Nottingham

2017 – 2021

B.Sc. (Hons) in Computer Science with Artificial Intelligence

First Class Honours

PUBLICATIONS

Conference Publications

1. **FlowPET: Physics-Informed Symplectic Flow Matching for Low-Count PET Reconstruction.**

Zheng Zhang, Hao Tang, Yingying Hu, Zhanli Hu, Jing Qin.

International Conference on Machine Learning (ICML), 2026. [CCF-A]

2. **FourierPET: Deep Fourier-based Unrolled Network for Low-count PET Reconstruction.**

Zheng Zhang, Hao Tang, Yingying Hu, Zhanli Hu, Jing Qin.

AAAI Conference on Artificial Intelligence (AAAI), 2026. [CCF-A, Oral Presentation]

Journal Publications

1. **Prompt-Agent-Driven Integration of Foundation Model Priors for Low-Count PET Reconstruction.**

X. Xie, W. Zhao, M. Nan, Zheng Zhang, Y. Wu, H. Zheng, D. Liang, M. Wang, Z. Hu.

IEEE Transactions on Medical Imaging (TMI), 2025. [JCR Q1, IF=9.8]

RESEARCH EXPERIENCE

SIAT, Chinese Academy of Sciences

2023 – 2025

Research Assistant

Advisor: Prof. Zhanli Hu

Foundation model priors and self-supervised disentanglement for low-count PET reconstruction.

Output: 1 JCR Q1 journal (TMI 2025).

ULIS, Imperial College London

2021 – 2022

M.Sc. Research Project

Advisor: Prof. Mengxing Tang

Neural super-resolution for ultrasound localization microscopy (ULM).

University of Nottingham

2020 – 2021

B.Sc. Final Year Project (Distinguished)

Advisor: Prof. Jianfeng Ren

Micro-Doppler radar sensing for human gait recognition.

AWARDS & HONOURS

• **Head's Scholarship**, University of Nottingham

2019 – 2020

Awarded to the top 10% of students in the department.

• **Distinguished Final Year Project Award**, University of Nottingham

2020 – 2021

Project: "Radar Human Activity Recognition Using Micro-Doppler Signature"